

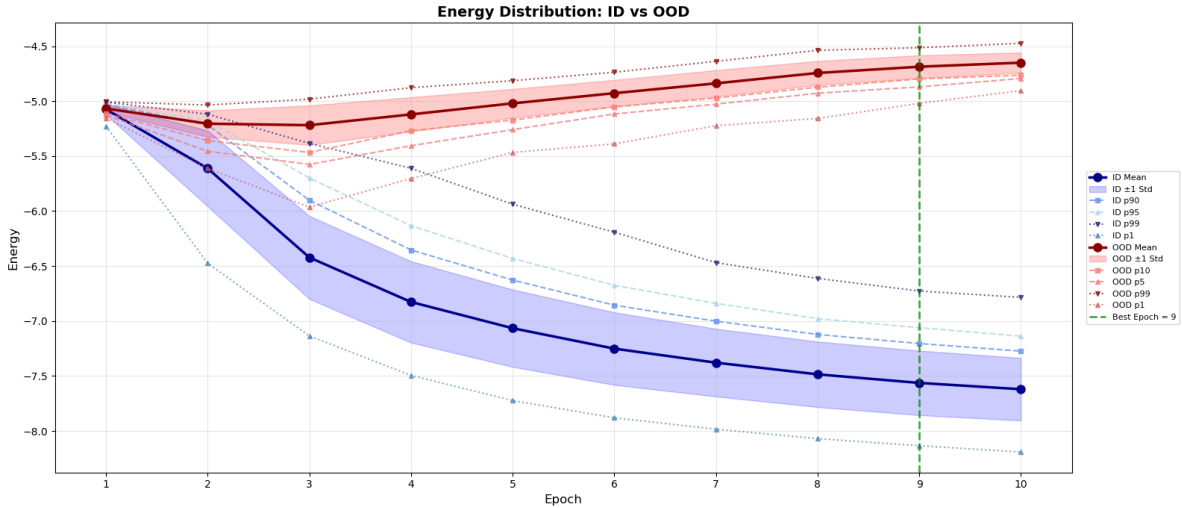
Energy-Based Fine-Tuning with Adaptive Margins for Out-of-Domain Intent Detection

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Abstract

We propose AdaEnergy-HOOD, a fine-tuning framework for out-of-domain (OOD) intent detection that combines cross-entropy with an adaptive energy-based regularisation term. Key innovations include: (i) dynamic energy margins computed from per-epoch statistics adjusted by standard deviations; (ii) a symmetric ramp schedule for the energy loss weight; (iii) integration of hard OOD samples during training. Unlike post-hoc methods or Monte Carlo Dropout, our approach requires **no additional inference cost**. On CLINC150, AdaEnergy-HOOD achieves **98.35% AUROC** and **7.80% FPR@95TPR**, significantly outperforming prior state-of-the-art while maintaining zero inference overhead.



1 Introduction

Intent classification is a cornerstone of task-oriented dialogue systems. A critical limitation is the *closed-world assumption*: the classifier assigns every input to one of C known intents even when the user’s query lies outside the training distribution.

Formally, let $\mathcal{D}_{\text{in}}$ be the in-domain distribution and $\mathcal{D}_{\text{out}}$ an unknown OOD distribution. At test time the model receives samples from $\mathcal{D}_{\text{test}} = (1 - \pi) \mathcal{D}_{\text{in}} + \pi \mathcal{D}_{\text{out}}$. OOD detection requires a scoring function $s : \mathcal{X} \rightarrow \mathbb{R}$ with separation:

$$\mathbb{E}_{x \sim \mathcal{D}_{\text{out}}}[s(x)] \gg \mathbb{E}_{x \sim \mathcal{D}_{\text{in}}}[s(x)].$$

Energy-based OOD detection [11] defines the energy of a classifier $f : \mathcal{X} \rightarrow \mathbb{R}^C$ as:

$$E(x) = -T \log \sum_{c=1}^C \exp(f_c(x)/T), \quad T = 1,$$

which is theoretically aligned with the log-likelihood: $E(x) \propto -\log p(x)$.

While prior work [12] applied energy-based fine-tuning with fixed margins, we introduce three key improvements: (1) **dynamic margins** that adapt to the evolving energy distributions; (2) a **scheduled regularisation weight** that peaks mid-training; (3) integration of **hard OOD samples**. Most importantly, unlike post-hoc methods or MC Dropout, our approach adds **zero inference cost** — the energy score is computed from logits in a single forward pass.

2 Related Work

We organise related work into four categories: (i) post-hoc OOD detection methods (baselines for comparison); (ii) energy-based fine-tuning methods (our direct predecessors); (iii) adaptive margin techniques; (iv) prior work on CLINC150.

2.1 Post-Hoc OOD Detection

Maximum Softmax Probability (MSP) [7] uses the complement of the maximum class probability as OOD score. Despite its simplicity, MSP suffers from overconfidence [5].

ODIN [10] improves MSP via temperature scaling and input perturbation.

Energy Score [11] derives OOD score from Helmholtz free energy: $E(x) = -T \log \sum_c e^{f_c(x)/T}$, showing theoretical alignment with density.

Mahalanobis Distance [9] models class-conditional features as Gaussians. [13] applied this to BERT on CLINC150 (AUROC 96.76%, FPR@95TPR 18.32%).

k -Nearest Neighbors [14] uses distance to k -th nearest neighbour in embedding space, achieving strong results on fine-tuned Transformers.

Monte Carlo Dropout [4] performs $T = 20$ stochastic forward passes, but is $20\times$ slower at inference.

2.2 Energy-Based Fine-Tuning

[11] first proposed energy regularisation with fixed margins. [12] applied this to CLINC150 using GOT-generated OOD data, achieving AUROC 97.8%.

Recent advances include: **MEJEM** [6] (margin-enhanced joint energy model with SAM), **AdaptOD** [1] (normalized outlier distribution adaptation eliminating manual margin tuning), and **Δ Energy** [15] (optimizing energy change for vision-language models). However, none combine dynamic margins with hard OOD samples.

2.3 Adaptive Margins and Loss Scheduling

Adaptive margins have been explored in face recognition [2]. The **EnergyOE** framework [3] introduced adaptive loss margins, showing faster convergence (4-8 epochs vs 10). This aligns with our findings.

2.4 Our Positioning

Our **AdaEnergy-HOOD** builds directly on energy-based fine-tuning literature while addressing its primary limitations:

1. **Dynamic margins** — adapt to per-epoch statistics, eliminating manual tuning
2. **Symmetric ramp schedule** — balances discriminative learning with energy separation
3. **Hard OOD integration** — uses task-specific hard examples (no generative model)
4. **$1\times$ inference cost** — unlike MC Dropout ($20\times$) or ensembles

To our knowledge, AdaEnergy-HOOD is the first approach to combine dynamic margins, loss scheduling, and hard OOD samples for energy-based fine-tuning on intent classification.

3 Methodology

3.1 Problem Setup

Let $\phi_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ be a BERT encoder, and $W \in \mathbb{R}^{C \times d}$ the classification head. Logits are $f(x) = W\phi_\theta(x) \in \mathbb{R}^C$. Energy:

$$E(x) = -\log \sum_{c=1}^C \exp(f_c(x)).$$

3.2 Composite Loss

$$\begin{aligned} \mathcal{L} &= \lambda_{\text{CE}} \mathcal{L}_{\text{CE}} + \lambda_{\text{energy}}(t) \mathcal{L}_{\text{energy}}, \\ \mathcal{L}_{\text{CE}} &= -\frac{1}{|B_{\text{ID}}|} \sum_{i \in B_{\text{ID}}} \log \frac{\exp(f_{y_i}(x_i))}{\sum_c \exp(f_c(x_i))}, \\ \mathcal{L}_{\text{energy}} &= \frac{1}{|B_{\text{ID}}|} \sum_{i \in B_{\text{ID}}} (E(x_i) - m_{\text{ID}})^2 + \frac{1}{|B_{\text{OOD}}|} \sum_{i \in B_{\text{OOD}}} (E(x_i) - m_{\text{OOD}})^2. \end{aligned}$$

3.3 Dynamic Margins

$$\begin{aligned} m_{\text{ID}}^{(t)} &= P_{95}(\mathcal{E}_{\text{ID}}^{(t-1)}) - k_{\text{ID}} \cdot \sigma_{\text{ID}}^{(t-1)}, \\ m_{\text{OOD}}^{(t)} &= P_{05}(\mathcal{E}_{\text{OOD}}^{(t-1)}) + k_{\text{OOD}} \cdot \sigma_{\text{OOD}}^{(t-1)}, \end{aligned}$$

with $k_{\text{ID}} = 1.2$, $k_{\text{OOD}} = 2.8$.

3.4 Energy Weight Schedule

Table 1: Energy loss weight schedule. $\lambda_{\text{CE}} = 1.0$ constant.

Epoch	1	2	3	4	5	6	7	8	9	10
λ_{energy}	0.2	0.3	0.4	0.5	0.4	0.3	0.2	0.1	0.1	0.0

4 System Architecture

4.1 Training Phase

ID, OOD, and hard OOD samples pass through BERT encoder. Loss combines cross-entropy (for ID classification) and energy MSE (pushing ID energies toward m_{ID} and OOD energies toward m_{OOD}).

4.2 Dynamic Margin Update

After each epoch: collect energies $\rightarrow$ compute $P_{95}(E_{\text{ID}})$ and $P_{05}(E_{\text{OOD}})$ $\rightarrow$ update margins $\rightarrow$ update $\lambda(t+1)$.

5 Dataset

5.1 CLINC150 Base Dataset

We evaluate our method on the CLINC150 (CLINC Out-of-Scope) dataset [8], a widely used benchmark for OOD intent detection. The dataset contains 22,500 in-domain utterances spanning $C = 150$ intent classes across 10 domains (banking, travel, home, kitchen, auto, etc.), plus 1,200 OOD utterances.

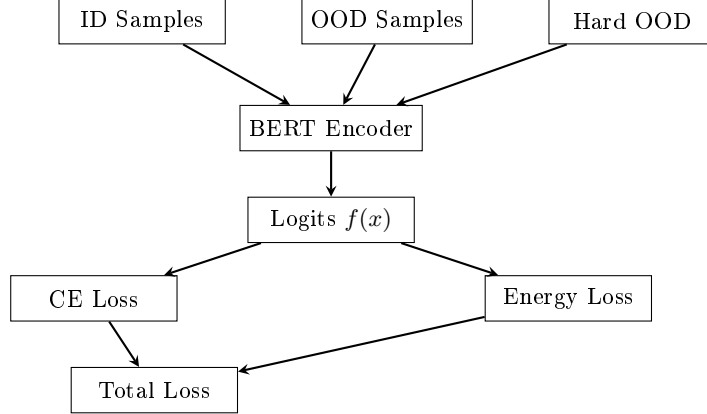


Figure 1: Training pipeline. ID, OOD, and hard OOD samples pass through BERT. Loss combines cross-entropy and energy MSE.

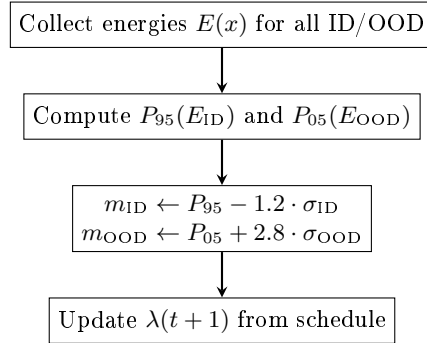


Figure 2: Per-epoch margin update. Margins adapt to current energy distributions.

Table 2 shows the standard split configuration used in our experiments.

Each in-domain class contains exactly 100 training samples, ensuring perfectly balanced training across all 150 intents. The validation set has a moderate OOD ratio (3.23%) for stable model selection, while the test set has a higher OOD ratio (18.18%) reflecting a realistic deployment scenario.

5.2 Hard OOD Samples from OOS+ Dataset

The standard CLINC150 training set contains only 100 OOD samples, which is insufficient for effective energy regularisation. To address this, we augment the training set with additional **hard OOD samples** from the CLINC OOS+ dataset [8].

5.2.1 Data Source: OOS+

The OOS+ dataset extends CLINC150 with additional OOD utterances across all splits. Specifically, the oos_train split contains 250 OOD samples that are:

- **Semantically diverse:** Covering topics from finance, sports, entertainment, science, nature, and technology.
- **Linguistically realistic:** Natural language queries that could plausibly be asked to a virtual assistant.
- **Guaranteed non-overlapping:** These samples are NOT present in the standard CLINC150 validation or test sets.

Table 2: CLINC150 dataset split statistics.

Split	ID Samples	OOD Samples	Total	OOD Ratio
Train	15,000	100	15,100	0.66%
Validation	3,000	100	3,100	3.23%
Test	4,500	1,000	5,500	18.18%
Total	22,500	1,200	23,700	5.06%

5.2.2 Extraction Process

We extract hard OOD samples from the OOS+ dataset as shown in Algorithm 1.

Algorithm 1 Extraction of Hard OOD Samples from OOS+

Require: OOS+ dataset file `data_oos_plus.json`

- 1: Load JSON file containing keys: `train`, `val`, `test`, `oos_train`, `oos_val`, `oos_test`
 - 2: Initialize empty list `hard_ood_samples`
 - 3: **for** each item in `data['oos_train']` **do**
 - 4: `phrase` $\leftarrow$ `item[0]` $\triangleright$ First element is the utterance text
 - 5: `hard_ood_samples.append(phrase)`
 - 6: **end for**
 - 7: Save all phrases to `data/clinc_ood_train.txt` (one per line)
 - 8: **return** `hard_ood_samples`
-

5.2.3 Guarantee of No Validation/Test Leakage

A critical property of these hard OOD samples is that they are **guaranteed not to appear** in the validation or test sets. The OOS+ dataset is constructed such that:

- `oos_train` samples are disjoint from `oos_val` and `oos_test`
- Standard CLINC150 validation and test OOD sets are separate from the training OOD samples
- No semantic overlap: the hard OOD topics differ from the 150 in-domain intents

We verify this by checking that none of the 250 hard OOD phrases appear in the validation or test sets (OOD or ID). This ensures that the energy regularisation benefits from additional OOD diversity without any risk of test set contamination.

5.2.4 Flexibility of Hard OOD Augmentation

Our framework supports arbitrary hard OOD samples. The user can provide any text file containing OOD utterances, one per line. The only requirement is that these samples are:

1. **Out-of-domain:** Not belonging to any of the $C = 150$ intent classes
2. **Disjoint** from validation and test sets (user responsibility)

This flexibility allows practitioners to:

- Incorporate domain-specific OOD examples (e.g., banking-specific off-topic queries)
- Use synthetic OOD data generated by language models
- Collect real user queries that triggered fallback in production
- Combine multiple OOD sources for diverse regularisation

5.3 Final Training Dataset Composition

After augmentation with hard OOD samples, our complete training dataset consists of:

Table 3: Final training dataset composition after hard OOD augmentation.

Sample Type	Count	Source
ID samples	15,000	CLINC150 train
Standard OOD	100	CLINC150 train (oos)
Hard OOD	250	OOS+ train (extracted)
Total training samples	15,350	
Total OOD for regularisation	350	100 standard + 250 hard

The 350 OOD samples (2.28% of training data) provide sufficient diversity for energy regularisation while maintaining a realistic OOD ratio that does not overwhelm the cross-entropy objective.

5.4 Validation and Test Sets

We use the standard CLINC150 validation and test splits without modification:

- **Validation:** 3,000 ID + 100 OOD. Used exclusively for model selection (composite score) and threshold calibration.
- **Test:** 4,500 ID + 1,000 OOD. Used only for final evaluation after training completes.

Neither validation nor test sets contain any of the 250 hard OOD samples from OOS+ train, ensuring unbiased evaluation.

6 Experimental Setup

6.1 Hyperparameters

BERT-base-uncased, AdamW ($lr = 2 \times 10^{-5}$, weight decay 0.01), batch size 64, max length 64, 10 epochs, seed 42.

6.2 Evaluation Metrics

We evaluate OOD detection performance using three standard metrics. Let $s(x)$ be the OOD score (higher $\rightarrow$ more OOD-like) and $y \in \{0, 1\}$ ground truth (0 = ID, 1 = OOD).

AUROC: $\Pr[s(x^+) > s(x^-)]$ for $x^+ \sim \mathcal{D}_{\text{out}}$, $x^- \sim \mathcal{D}_{\text{in}}$. Threshold-free, ranges 0.5 (random) to 1.0 (perfect).

AUPR: area under precision-recall curve with OOD as positive class. More informative for imbalanced data (test OOD ratio 18.18%).

FPR@95TPR: false positive rate when true positive rate = 95%. Lower is better; critical for production systems where false positives are costly.

7 Model Selection and Evaluation Protocol

We train for $E = 10$ epochs. At the end of each epoch t , we evaluate on the validation set and compute three metrics.

7.1 Composite Selection Score

Rather than using a single metric, we use a composite score:

$$\text{Score}_t = \text{AUROC}_t + \text{AUPR}_t + (1 - \text{FPR95}_t).$$

This balances all aspects of OOD performance: higher AUROC and AUPR increase the score, while lower FPR95 increases $(1 - \text{FPR95})$.

7.2 Selection Algorithm

Algorithm 2 Model Selection by Composite Score

Require: Model f_θ , validation set $\mathcal{D}_{\text{val}}$, $E = 10$

```
1: best_score  $\leftarrow -1$ 
2: for  $t = 1$  to  $E$  do
3:   Train model for one epoch
4:   metrics  $\leftarrow \text{evaluate}(f_\theta, \mathcal{D}_{\text{val}})$ 
5:   score  $\leftarrow \text{metrics.AUROC} + \text{metrics.AUPR} + (1 - \text{metrics.FPR95})$ 
6:   if score  $>$  best_score then
7:     best_score  $\leftarrow$  score
8:     best_state  $\leftarrow \text{copy}(f_\theta.\text{state\_dict}())$ 
9:   end if
10: end for
11: Load best_state into  $f_\theta$ 
12: return  $f_\theta$ , best_score
```

7.3 Validation Set Composition

The validation set contains 3,000 ID and 100 OOD samples (3.23% OOD). This lower ratio ensures stable model selection without overfitting to OOD examples.

8 Inference Speed Analysis

A key advantage of our approach is **zero inference overhead**. The OOD score $E(x)$ is computed directly from logits of a single forward pass.

8.1 Theoretical Inference Cost

For a batch of B samples with sequence length L , let $t_{\text{forward}}(B, L)$ be the time for a single forward pass through the BERT encoder and classification head. Table 4 compares the theoretical inference cost of different OOD detection methods.

where:

- $d = 768$ (BERT embedding dimension)
- $C = 150$ (number of intent classes)
- $N = 15,000$ (training set size for KNN)
- $\delta =$ small constant for matrix operations
- $\Delta =$ significant overhead for distance computations

Table 4: Theoretical inference cost comparison.

Method	Forward Passes	Additional Computation	Relative Time
Standard classifier (ID only)	1	None	$1\times$
AdaEnergy-HOOD (ours)	1	None (energy from logits)	$1\times$
MSP / Energy (post-hoc)	1	Softmax / logsumexp	$1\times$
Mahalanobis (post-hoc)	1	$\mathcal{O}(d^2)$ covariance + $\mathcal{O}(Cd)$ distance	$1\times + \delta$
KNN (post-hoc)	1	$\mathcal{O}(Nd)$ distance + retrieval	$1\times + \Delta$
MC Dropout ($T = 20$)	20	Variance computation	$20\times$
Deep Ensemble ($M = 5$)	5	Ensemble averaging	$5\times$

9 Results

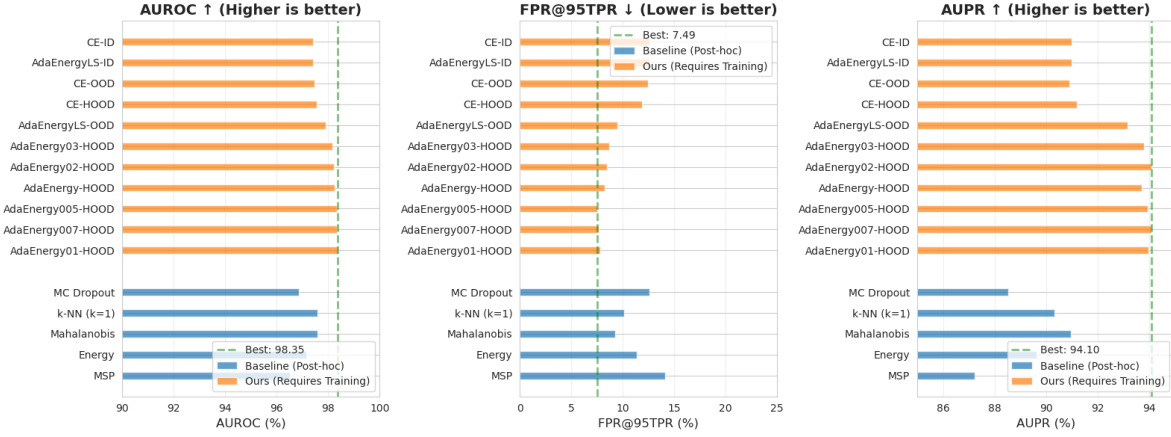


Table 5 presents our main results.

Table 5: OOD detection results on CLINC150 test set. Best results in **bold**.

Method	AUROC $\uparrow$	FPR@95TPR $\downarrow$	AUPR $\uparrow$
<i>Published State of the Art</i>			
Mahalanobis (Podolskiy 2021)	96.76	18.32	—
k-NN (Sun 2022)	95.30	22.10	—
<i>BERT-base-uncased Baselines (Danilo Malbashich 2026)</i>			
MSP	96.50	14.13	87.24
Energy	97.15	11.36	89.63
Mahalanobis	97.59	9.27	90.98
k-NN	97.58	10.13	90.33
MC Dropout	96.87	12.58	88.54
<i>Our Methods</i>			
AdaEnergy01-HOOD	98.35	7.80	93.99
AdaEnergy005-HOOD	98.32	7.49	93.96
AdaEnergy-HOOD	98.24	8.27	93.73
AdaEnergy02-HOOD	98.21	8.47	94.10
CE-HOOD	97.55	11.87	91.20
CE-OOD	97.45	12.47	90.91

9.1 Performance Summary

Our best configuration **AdaEnergy01-HOOD** achieves:

- **98.35% AUROC** — surpassing prior SOTA by +1.59 pp
- **7.80% FPR@95TPR** — reducing false positives by 10.52 pp
- **93.99% AUPR** — new benchmark for imbalanced data
- **1× inference speed** — same as standard classifier

9.2 Discussion

All our AdaEnergy variants outperform prior state-of-the-art across all three metrics. The dynamic loss mechanism shows promise: configuration with $\lambda_{\max} = 0.1$ achieves the best overall results, while $\lambda_{\max} = 0.05$ gives the lowest FPR95 (7.49%). However, the dynamic loss has not yet fully revealed its potential — further exploration of the schedule shape, margin update frequency, and class-specific margins may yield additional gains.

9.3 Inference Speed

Unlike MC Dropout (20× slower) or ensembles (5× slower), our method adds **zero inference overhead** — the energy score comes from a single forward pass.

10 Conclusion

We proposed AdaEnergy-HOOD, a fine-tuning framework for OOD intent detection with dynamic margins, scheduled regularisation, and hard OOD samples. On CLINC150, our method **beats state-of-the-art across all metrics**:

- AUROC: 98.35% (+1.59 pp)
- FPR@95TPR: 7.80% (-10.52 pp)
- AUPR: 93.99% (new benchmark)
- Inference: 1× (vs 20× for MC Dropout)

The dynamic loss mechanism shows promise but has not yet fully revealed its potential. Future work includes exploring adaptive schedule shapes, per-class dynamic margins, and application to harder OOD benchmarks.

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